CHEAT SHEET · DAY 2

Data management II

Combining files, recoding, tables and charts for a pack. Every command here was taught today; the deck shows each one at work on the LFS or the CPI.

Building the quarter — Part 1

`use "data/jan.dta", clear`	the first month in memory
`append using "f2.dta" "f3.dta", generate(source)`	add the rows of more files
`tabulate source`	which file each row came from
`describe, short`	rows and variables in four lines
`assert _N == 5888`	stop if the row count is wrong
`save "data/q1.dta", replace`	keep the quarter

Merge — Part 2

`isid key`	is the key unique in this file?
`merge m:1 key using "f.dta"`	many rows in memory, one in the file
`merge 1:1 key using "f.dta"`	one row each side
`..., assert(match)`	stop unless every row matches
`..., nogenerate`	do not create _merge
`..., keepusing(v1 v2)`	bring only these variables
`tabulate _merge`	1 master only, 2 using only, 3 both
`never merge m:m`	find the key that makes one side unique

Reshape — Part 3

`reshape wide v, i(row) j(col)`	long to wide: values of col become suffixes
`reshape long v, i(row) j(col)`	wide to long: suffixes become values
`egen t = rowtotal(v1 v2 v3)`	a total across columns, missing as 0
`replace v1 = 0 if missing(v1)`	only when empty means zero

Tables — Part 4

`tabulate v`	counts and shares of one variable
`tabulate r c`	two-way counts
`tabulate r c, missing`	show the missing as a column
`tabulate r c, row nofreq`	shares within each row
`tabulate r c, column nofreq`	shares within each column
`summarize v, detail`	percentiles, mean, spread, extremes
`tabstat v, statistics(mean p50 n) by(g)`	statistics by group
`table (r) (c)`	release table of counts
`table (r) (c), statistic(percent, across(c))`	row percentages
`table ..., statistic(mean v) statistic(count v)`	mean with its count
`..., nformat(%5.1f)`	one decimal

Aggregates — Part 5

`preserve ... restore`	work on a copy, then get the persons back
`collapse (mean) v (count) n=v, by(g)`	one row per group
`contract a b, freq(n) nomiss`	counts of combinations, as data
`egen s = mean(v), by(g)`	the group value on every row
`egen r = rank(-v)`	rank, highest = 1

Graphs — Part 6

`graph hbar (mean) v, over(g)`	bars across, one per group
`over(g, sort(1) descending)`	longest bar first
`title() ytitle() note()`	what, unit, source
`histogram v [, frequency]`	the shape of one variable
`twoway scatter y x [, mlabel(g)]`	one point per row
`graph export "output/f.png", replace`	save the chart on screen

Export — Part 7

`collect export "f.xlsx", replace`	the last table to Excel (or .docx)
`putexcel set "f.xlsx", sheet("S") replace`	choose file and sheet
`putexcel A1 = "text" B1 = (r(mean))`	text and results into cells
`return list`	what the last command stored
`putdocx begin / putdocx save "f.docx", replace`	open and save a Word document
`putdocx paragraph / text ("...") / image "f.png"`	content
`putdocx collect`	the last table into Word
`putdocx clear`	close a document left open

Loops — Part 8

`local name "text"`	a name for a piece of text
`` `name' ``	use it: backtick, name, apostrophe
`display "`name'"`	see what it holds
`foreach v of varlist a b c {`	repeat for each variable
`levelsof g, local(L)`	the values of g into a local
`foreach x of local L {`	repeat for each value
`}`	alone on the last line

Help inside Stata: `help` *command*, and the PDF manual linked at the top of each help page.